



Constructability Review Form: M6-PM-2012 - Eversource

Substation Constructability Review

Project Title: Hyde Park Battery Energy Storage System		☐	Prior Constructability Review Performed?
		☐	Program Related Site-Specific Review?
Project Numbers(s): 21334		Project Constructability Phase:	
Date(s) of Constructability Review: 10/31/2023		☒ Conceptual Engineering Design	
07/20/2023		☐ Preliminary Engineering Design	
		☐ Detailed Engineering Design	
Project Scope Statement: Installation of a 10MW/20MWh utility-scale battery storage system (BESS) along Cummings Highway near Hyde Park Substation #496 in Hyde Park, MA. This constructability review is being based off an initial project walkdown and tabletop review and is being completed prior to Conceptual Engineering for the BESS development. More encompassing reviews will be completed as the project progresses through engineering stages.			

Department Review/Walkdown Attendees*:

Required [X]	Attended [X]	Department	Name	Required [X]	Attended [X]	Department	Name
☐	☒	Project Management	M. Rosenberg	☐	☒	S/S Engineering	P. Fitzgerald
☐	☐	Asset Management		☐	☒	P&C Engineering	T. Callahan
☐	☒	Constr. Management	R. Taves, J. Didomenico	☐	☐	Capital Projects Eng.	L. Croteau, S. Horner
☐	☐	S/S Technical / LCE		☐	☒	Telecom	F. Tahmasbi
☐	☐	S/S Operations**		☐	☒	Environmental	C. Newhall, C. Russell
☐	☐	Field Engineering		☐	☐	Real Estate	
☐	☐	Line Engineering		☐	☐	ROW T&D Rights & Survey	
☐	☐	Vegetation Management		☐	☐	Survey Engineering	
☐	☒	Siting	Leah Zambernardi	☐	☐	Cost Estimating	
☐	☐	Safety		☐	☒	Project Services	Meredith Boericke
☐	☐	SCADA		☐	☐	Material Procurement	

* Sign-off in the table above for the Detailed Engineering Design Constructability Review indicates that all Action Items/Concerns from prior Constructability Reviews have been mitigated/resolved as documented in this form. Not every department is required to attend walkdowns for every project but should be consulted for project impact considerations.

** Internal Eversource Substation Operations must be included in the review even if external construction resources are planned

[Project Manager] certifies that this review was performed to completeness, that all applicable departments were informed of the project and have had opportunity to provide input to this review:

Michael Rosenberg
Name / Signature

12/20/23

Date



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Documents available for use prior to Constructability Review					
[X]	Document	[X]	Document	[X]	Document
☒	Initial Scope -Birdseye site plan -General Arrangement	☐	Preliminary Drawings -Civil Site Drawing -Electrical Drawing -Section Drawings	☐	Prints in hand for marking up -Panel Front/Read View -Control Enclosure Layout -Floor / Wall Plans
☐	Detailed Scope Document	☒	Proposed Schedule	☒	Construction Sequence/Timeline
☐	Previous Notes / Comments / Observations	☐	Major Material / Equipment List	☐	Replace / Reuse of equipment

Action Items and Resolutions:

Item No.	Const. Review Action Item/Concern	Design Resolution/Mitigation:
1	Underground obstructions and conditions to be determined through required soil borings. It is understood that the site is a former landfill for construction debris and efforts will be made during the design process to limit excavation when possible	Through engineering stages, utilize design options to limit amount and depth of excavation.
2	Views of BESS from abutting neighborhood and concerns for losing open land could be mitigated by scope/design alterations. Visual screening will likely be required, siting BESS closer to shopping plaza & away from neighborhoods. Screening/buffering likely required for abutting residential neighborhood.	Screening, plantings, and orientation of BESS and its affect on the neighborhood will be a focus of the vendor design.
3		
4		
5		

	Key Review Item Description	N/A or No Concern	Review Notes, Concerns, or Additional Info Needed
A	Engineering & Asset Management		



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1	Scope of Work - Is scope adequately developed and sufficiently detailed for intended deliverable? - Any complimentary projects or other adjacent obsolete equipment for inclusion? - Any overlapping projects that could impact work/outages? - Consider Physical Installation, P&C Scope, Civil Scope, Engineering studies, Design standard adherence - Any remote station impacts? - Engineering Estimates for Design and Technical Support? (EE and EOC) Total Project	☐	Scope of work has been completed for both internal Eversource engineering and external Battery Energy Storage System vendor. Next steps following partial funding will be to progress engineering deliverables up to Preliminary Engineering where many of the items in the section will be addressed.
2	Materials Identified/Ordered -Any concerns with BOM/Major Materials? Review BOM lead times and project need dates -Specialty Major Materials: Identify Specialty Equipment & Materials potentials - temporary/replacement - long lead items? -Any Materials needed to facilitate construction access?	☐	Major material lead times for batteries, grounding transformers, and inverters are estimated to be at 12 months +/- . This duration has been built into the project schedule including standard procurement practices based on funding schedule.
3	Control House - Review Unmapped equipment interferences, Control/Battery Enclosure Layout, HVAC System (if equipment added), space for new or modified equipment, AC / DC station breaker availability and service loading, Panel Space, Expansion, one-line and/or riser type panelboard drawings, other.	☐	Indicative pricing effort has provided us with multiple equipment layouts for the site. Further engineering will be required to finalize layout, but the site is able to accommodate the installation of the equipment in multiple locations.
4	Ratings & Dimensions for Major Equipment. Any concerns upon field review? Equipment Rating One Line (and/or equipment thermal ratings tables) - Review for any errors in field	☒	No site concerns. As engineering progresses, we will need to make sure that the new access road will be able to handle the equipment.



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5	Wall, Fence, Gate, and Foundation Considerations -Any permanent / temporary alterations or relocations required? -Pre-Project Sound Measurements Needed? Sound Walls? -Take measurements for Foundation quantities, dimensions and concrete volumes -CIP Review/Ballistic needs. Any Physical Security Requirement Consideration? -Assess condition of existing yard equipment foundations and viability for reuse (if proposed) -Review station expansion areas (if needed) for structures, old foundations, etc. -Any Steel Structural Plan Considerations? -Any impacts to construction plan?	☐	Site is currently fenced, and future Battery Energy Storage System and associated equipment will also be fenced. Sound studies will be completed further along in the engineering process to ensure system adheres to dB requirements.
6	Below Grade / Enclosure Raceway - If possible, evaluate existing raceway and review spacing requirements. Review for any subsurface investigation needs or potential below grade obstructions. GPR/test pits needed? Geotechnical Investigation - is soil boring or resistivity investigation needed?	☐	See item 7 below.
7	Underground Obstructions -Drawing review concerns? -Is preconstruction or test excavation needed? -Is reuse of underground control cables / Conduits /replace Non-Shielded Cable feasible (if applicable) -Are spare conduits available? Are they useable? -Review if any areas will need to be plated for increased load bearing for equipment access (i.e., large Xfmr, etc.)	☐	Underground obstructions and conditions to be determined through required soil borings. It is understood that the site is a former landfill for construction debris and efforts will be made during the design process to limit excavation when possible.
8	Grounding Plan (Ground Grid) - Will footprint impact layout or other equipment? Identify locations to improve grounding. Impacts to foundations and ground grid plans. Observations on wetlands or ability to perform tests	☐	This is a new location, so a new ground grid will be installed in the location of the BESS. Coordination with the foundation plan and other equipment will be done to ensure the proper ground grid size. No site conditions pose any challenges for the ground grid.



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9	Nomenclature / Switching One-Line Diagram -Any obvious deviations from diagrams provided?	☒	No
10	BPS Separation Requirements - Is this BPS Substation? Y/N (part of PAF) - Changes to pre-existing BPS standards?	☒	No
11	IEC 61850 Substation Automation and Communication Requirements - Any Impacts/Retrofit needed?	☐	No
12	Any Upgrades to ES Standards - Is retrofit needed to bring up to ES standards? Can that be done?	☐	Capital Projects Engineering has written technical specifications that will be utilized in the Battery Energy Storage System Request for Proposal. These technical specifications are being reviewed and cross referenced with a draft document that the Eversource Standards group has drafted.
B	Construction		
1	Any safety hazards specific to the facility? i.e., Low Bus, Deterioration, etc.	☐	No
2	Prior known Site-Specific Conditions -Common or Obvious station Issues -Missing scope items that could impact costs?	☐	Unknown subsurface obstructions being carried as a risk until soil boring are completed.
3	Identify/Consider Required Depts for Construction? -Who is constructor: Internal vs External Resources -S/S Ops, GO, Test, Field Engineering, etc. -Specialty Contractors Required? -Required Oversight Needed - Estimated level of CR and LCE support needed	☐	Outside vendor will complete Construction of Battery Energy Storage System. Civil work for underground distribution circuit extension to be completed by outside vendor while cable pulling and termination to be completed by internal resources.



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4	Site Plan, Access, and Temporary Construction Facilities: -Confirm maintenance access for proposed new or replaced equipment -Verify clear footprint for yard equipment / Property Plan -General access to site, can areas outside substation be utilized? -Substation Entry Plan (driveway and drive path modifications) - Identification of Unique Requirements, i.e. (access bridges, station service, etc.) -Site Development Plan (grading and drainage) with Cut/Fill -Existing Rock Conditions? Potential Impacts (rock quantity)? -Any police details needed for deliveries, restricted hours, and/or need for extended hour approvals -Any Sanitary Civil Requirements? (Septic tanks, etc.) -Review turning radius for delivery trucks (if critical) -Any Substation Access and site/yard access Limitations? -Office Trailer Needs: Facilities, Utilities, Restroom, Container, Dumpster Space -Weather impacts to Site (Snow Placement) -Identify Material Laydown and Security -Where will new vs removed equipment go? -Construction permit for trailers outside substation fence?	☐	Layout requirements have accommodated for both maintenance clearances between equipment as well as around equipment and fence perimeter. Existing access road will be improved and utilized to access location. Further site and grading plans to be developed as engineering progresses as well as a pre-construction plan that will address the other items.
5	Unique Equipment Considerations: Gas insulated substation, Gas insulated line, Gas insulated bus, Pass Breaker, Statcom, Sync Condenser	☒	This is Eversource's second BESS installation, and this should still be considered a piece of unique equipment to be installed.



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6	Outage & Energization Plan and Construction Sequence: Review electrical working clearances for equipment during construction, Detailed look at protection zone changes, Locked sequence or random outages, Outage durations minimized. Considerations: -Approach Distances – Outage Requirements to maintain electrical clearances throughout proposed installation -Distribution Clearances -Clearance Holder Assignments -Drawings vs Field Conditions -Outage boundaries and interconnections -Phased approach needs for additional mobilizations -Verify outage sequence supports the planned construction sequence and timeline. (Lead Commissioning Engineer "LCE", Substation Operations, and Outage Planner input) -Any design alterations or considerations to Eliminate SCLL Conditions, Reduce Recall Time, Reduce Outage durations, Special Conditions -Additional outage requirements: Railroad Catenaries, Fiber outage, SCADA outage, Relay permissions -Seasonal or weather conditions impacts	☐	Will be completed further along in the engineering process.
7	Equipment Testing & Acceptance -Grounding plans / Resources? Will we need grounds hung? -Hi-pot testing and where will the equipment be set up?	☐	Will be completed further along in the engineering process.
8	Need for mobile generator and where connected (if needed)	☒	Mobile generators not required for construction.
9	Need for mobile battery and where connected (if needed)	☒	Mobile batteries not required for construction.
10	OT work required or Notable Permit Conditions (special hours)?	☐	TBD
11	Coordination of boundaries/seams (Identification of point of demarcation for connection to existing equipment and/or for coordination with other municipalities)	☐	Will be completed further along in the engineering process.



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C	Environmental		
1	Soil Pre-Characterization or Grab Sampling Completed/Planned	☐	Soil borings and characterization planned.
2	Desktop investigation and EAD Review -Required Permits -Stormwater Considerations -Dewatering requirements -Wetland and buffer review	☐	EAD completed. Below are known permitting requirements at this time. -DEP Waste Management Program Post-Closure Use Permit for authorization to construct on closed landfill facility. -If soil is to be disposed offsite from Eversource property, soil samples will need to be taken to characterize the soil and a soil management plan will need to be developed to get approval from a disposal facility or landfill. - Requires annual Tier II reporting to MassDEP for batteries. - If project is expected to disturb ≥ 1 acre of land, then will require submittal of eNOI for coverage under the EPA 2022 Construction General Permit, incl. development of a SWPPP and associated inspections are required.
3	Risk or presence of PCB's/Asbestos/Lead/SF6	☐	Possible risk. Additional communication with DEP required along with review of landfill closure documents.
4	Transformer Spill Containment Recommended?	☐	Required spill containment will be determined with further engineering.
5	Expanding Footprint / New Buildings or future Expansion Plans? (Enviro Impact)	☐	Equipment will be ground mounted. A building will not be constructed.
6	Environmental oversight considerations? (Inspection type/Interval)	☐	Oversight will be required but length and frequency will be determined at permit application and approval stages.
7	Environmental mitigations - Any matting or seasonal needs? Review D&M Plan with CR.	☒	Not anticipated at this time.
D	Siting, Project Services, and ROW T&D Rights & Survey		
1	Any regulatory concerns or approvals needed? - Any concerns noted that could be mitigated by scope/design alterations? - Likely to require visual screening? What considerations (feasibility or location)? -Zoning Relief and Permits required?	☐	Views of BESS from abutting neighborhood and concerns for losing open land could be mitigated by scope/design alterations. Visual screening will likely be required, siting BESS closer to shopping plaza & away from neighborhoods. Screening/buffering likely required for abutting residential neighborhood. Zoning Relief and Permits are required. City of Boston Article 80 & Zoning Variance.
2	Other siting considerations?	☐	Neighborhood is concerned about losing open land. Also interested in green energy and energy savings solutions.



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3	Neighbor/Abutter impacts: - Any noise, visual impacts, or sensitive abutters - Any changes to lighting or additional lighting needed?	☐	Visual impact of equipment to be reviewed with City of Boston and community and accommodations to be made during engineering phase for any required screening of faux façade.
4	Review for roadway, gate access, trailers, equip height, noise, weekend/night work, or any potential public disturbance related to work	☐	Increased traffic will be seen during construction. Post construction, minimal traffic can be expected to the location for routine maintenance purposes.
5	Installation within Eversource Rights - Is Rights and Restrictions Review required?	☐	Eversource is owner of property
6	Expanding Footprint / New Buildings / Fence line or future Expansion Plans?	☐	Location of Battery Energy Storage System to take into consideration future development of the parcel for other electrical needs.
7	Proposed Equipment on same property? - Any Encroachments	☐	No encroachments anticipated
F	Protection & Controls		
1	Panel Graphic additions and/or Modifications	☐	Not applicable
2	Space on existing panels for new equipment? (front & rear of panel fronts)	☐	Not applicable
3	Panel fronts conform to drawings	☐	Not applicable
4	Access to bring new panels into control enclosure	☐	Not applicable
5	Nameplate list and location of nameplates	☐	Will be provided as engineering progresses
6	Verifies that relays and meters that will be an interconnected project work are as shown on the drawings.	☐	As engineering progresses drawings will be reviewed to ensure proper relaying and autotransfer scheme is achieved to allow the BESS to be supplying energy
7	Temporary protection, relays, control cable needed? How long will they be required?	☐	None anticipated at this time.
8	Will relay setting changes be needed? Any additional outage requirements for changes?	☐	At this stage of the engineering, it is not a requirement to do any additional relay setting changes as this will be interconnecting onto an existing distribution circuit. As the engineering progresses, the impact of the BESS interconnection on existing protection schemes will need to be assessed to determine the need for any remote setting changes.
G	Telecom Engineering		
1	Review telecom equipment in existing control enclosure	☐	Not Applicable



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2	Review path to get telecom circuits into the station and paths within the control house. Determine whether new underground duct or overhead tray work is needed.	☒	Fiber exists at the station and ADSS is available at 427 Cummins Highway if required. Wired or wireless options will be reviewed during the engineering phases.
3	Review current or proposed Telecom equipment power connections and validate with P&C any need for system 1 and system 2 separation.	☐	No need for system separation
4	Review current Telecom fiber diversity and validate with P&C any fiber diversity requirements.	☐	
5	Review and list any remote station impacts/updates	☐	Not Applicable
6	Confirm whether correct interface equipment is available (i.e.) for fiber termination or multiplexor termination	☐	This will be a new installation, and equipment will be specified to be installed
7	Confirm whether existing communications cabinets are enough to support additional telecommunications equipment installed	☐	New installation, will require new cabinets
8	Review space for any required additional communications cabinets which may be required to support additional telecommunications equipment installed	☐	New installation, will require new cabinets
9	Review third party Telecommunications at substation and confirm whether additional capacity or Ground Potential Rise protection is required	☐	

Photos

(Ex. Equipment control cabinet, Breakers, MODs, CCVTs, PTs, Transformers)